

Line Outage Detection Using Spatial Characteristics of The Power System From a Neural Network View

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Outline

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- Future Work

Background



- Modern power grids are seeing a **major increase** in renewable energy sources.
- In addition, the power grid is no longer static and tends to be dynamic.
- Renewable energy sources are **less predictable** and may inject reactive power into the grid, which can cause **instabilities**.
- As a result, the power grid becomes **more sensitive to grid events**, such as line outages.

Background



- As larger the power grid goes, the classic computations have **higher computational cost**.
- Outages, mitigations and other power grid events **aren't always reported** to the administrator due to various reasons.

Project Goal

Developing a data-driven solution using Graph Neural Networks to **quickly detect single / multiple line outages** (as a test case) using **the spatial characteristics of the grid**.

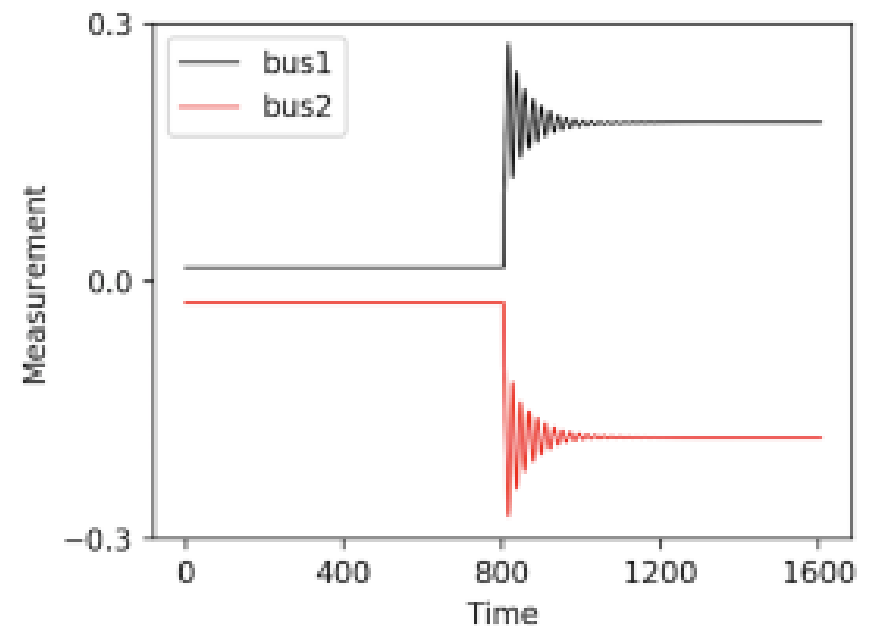
Literature Survey

He & Cheng (2020): Graph Convolutional Neural Networks for Power Line Outage Identification.

- Proposed a Graph Convolutional Network (GCN) to **identify which transmission line has failed** using bus phase measurements.
- GCNs can **accurately localize outages** using only measurement data, without requiring detailed system models or parameters as traditional methods do.

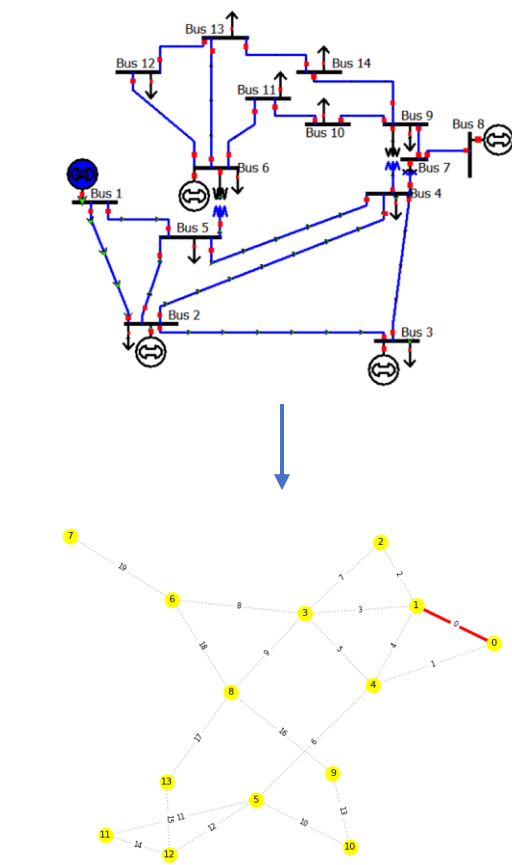
Literature Survey – Data

- Simulated line outage cases using the **Power System Analysis Toolbox (PSAT)**
- Collected phase measurements of bus phasor angle at each bus node.



Literature Survey – Data

- The power grid can be naturally modeled as a **graph**, where:
 - **Nodes** = buses (generators, loads, substations)
 - **Edges** = transmission lines
- This transformation enables the use of **GNNs** to learn and generalize from the grid's structure and behavior.



Literature Survey – Data Preprocess

Since the goal is to detect **sudden changes** caused by line outages, the paper explored ways to transform the data into the **frequency domain**:

1. Discrete Fourier Transform (DFT)

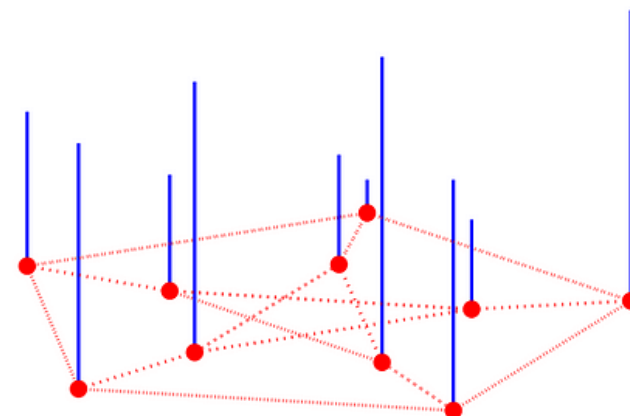
- Applied to the **time series** measurements.

2. Graph Fourier Transform (GFT) – the **more successful** option

- Applied to the **spatial distribution** across the grid.

GFT

- Just like the regular Fourier Transform breaks down signals into sine and cosine waves, the **Graph Fourier Transform** breaks down graph signals into **graph frequency components**.
- It helps reveal how **smooth** or **abrupt** a signal (like voltage angles) is across the structure of a graph.



GFT

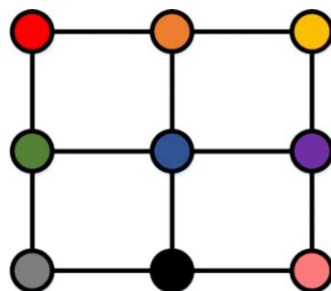
How does it work?

The Graph Fourier Transform is based on the **spectral decomposition of the normalized adjacency matrix**, which reveals how a signal (like voltage angles) varies across the network's structure.

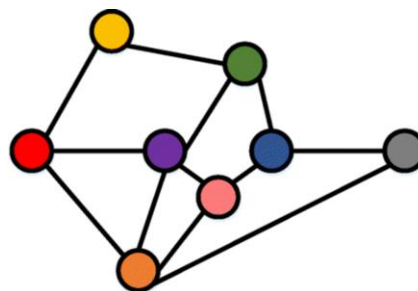
$$\text{DFT} = \frac{1}{\sqrt{N}} \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 1 & \omega & \cdots & \omega^{N-1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^k & \cdots & \omega^{k(N-1)} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{N-1} & \cdots & \omega^{(N-1)(N-1)} \end{bmatrix} \quad \hat{s} = \text{DFT}s \longrightarrow \tilde{A} = \text{GFT}^{-1} \Lambda \text{GFT} \quad \hat{s} = \text{GFT}s$$

GNNs

- GNNs operate similarly to CNNs - but while CNNs work on **grid-like data** (e.g., images), GNNs are designed to work on **graph-structured data**.
- Like CNNs, GNNs use local kernels to extract features - but instead of operating on fixed neighbors (like adjacent pixels), GNNs operate on a **node's neighbors in the graph**.



CNN



GNN

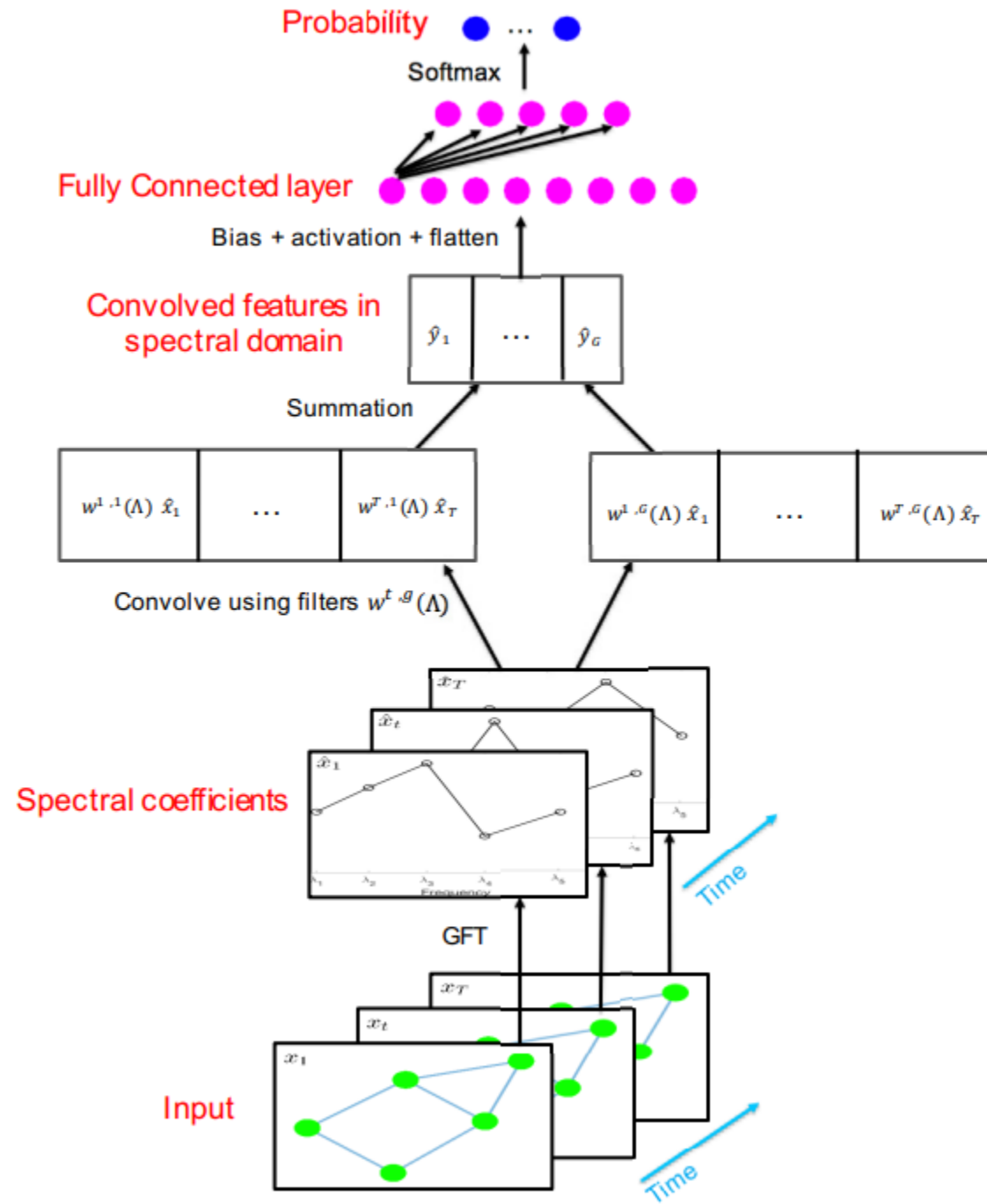
GNNs

- By defining the kernel as a **polynomial of the adjacency matrix** (or Laplacian), GNNs can capture information from **multi-hop neighborhoods** - e.g., a degree-H polynomial reaches H-hop neighbors.

$$y = \sum_{i=0}^H (\omega_i \tilde{A}^i) x$$

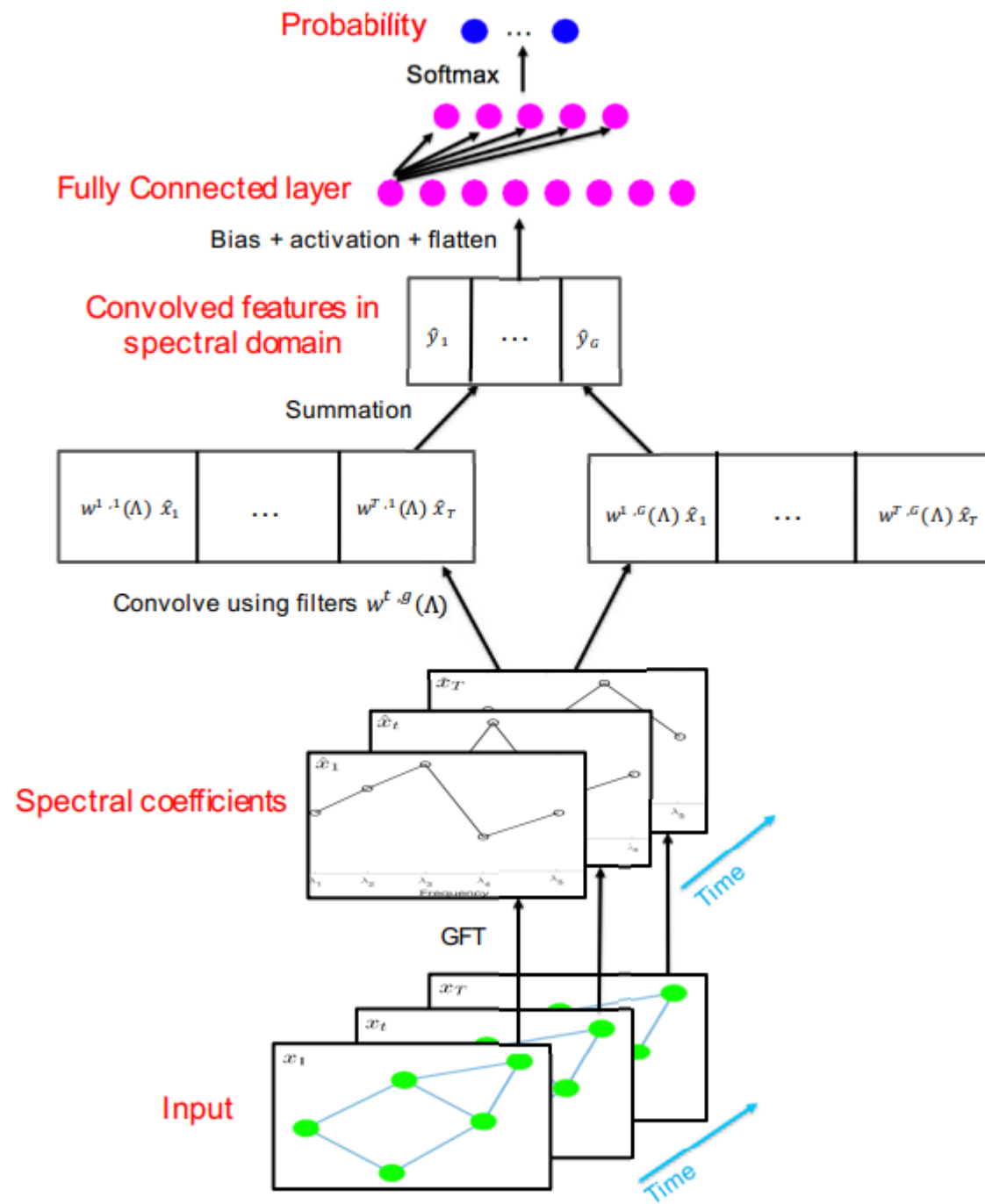
Literature Survey - architecture

1. Collect time-series data
2. Model the power grid as a graph
3. Apply Graph Fourier Transform
4. Feed transformed features into a GNN
5. Apply softmax layer



Literature Survey – architecture

Since the model is built to predict the **probability of outage** for each power line, detecting **multiple line outages** do not incur additional computational complexity or cost.



Dataset Overview

- **Goal:** Generate **realistic measurements for line outage identification**.
- **Networks used:** IEEE test cases (14, 39 and more).
- **Simulator:** Pandapower — open-source power system analysis toolbox.
- **Outputs:** Time series of **bus phasor angles** + outage labels.

PandaPower

- Open-source Python tool for **power system analysis**.
- Models grids as networks of buses (nodes) and lines/transformers (edges)
- Provides:
 - **Load-flow calculations** (steady-state analysis)
 - **Short-circuit analysis**
 - **Optimal power flow**
- Easy interface with **NumPy/Pandas**.

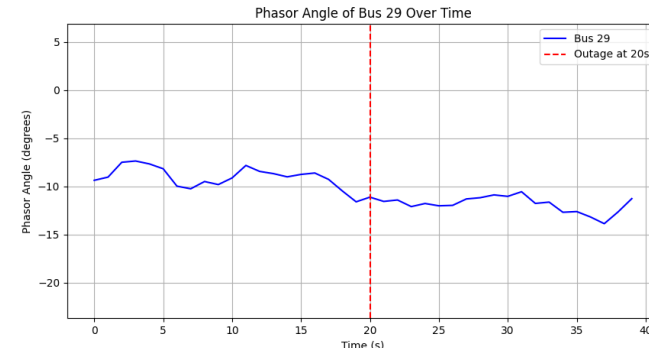
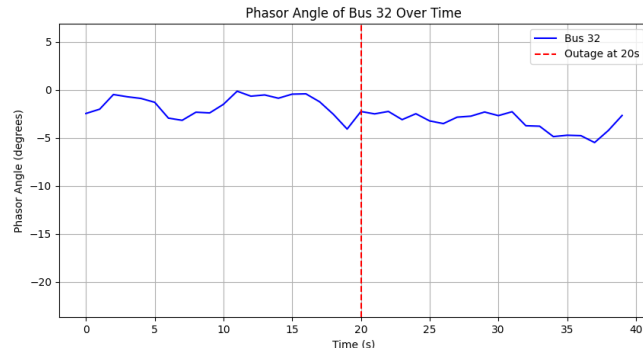


How the data is generated

- Start from a standard IEEE test network.
- For each outage case:
 - Choose 0, 1, or 2 lines to disconnect.
 - Run **steady-state power flow** with PandaPower.
 - Introduce the outage at a fix step.
- Store the resulting bus angle values, the corresponding outage label and the **graph topology**.

Noising the Data

- Motivation: **Real PMU data contains fluctuations** even without faults.
- **Noise injection:**
 - Small fluctuations in **bus loads** (both active and reactive).
 - Creates small variations in voltage angles even in “normal” cases.
- This ensures the dataset mimics **realistic operating conditions**.

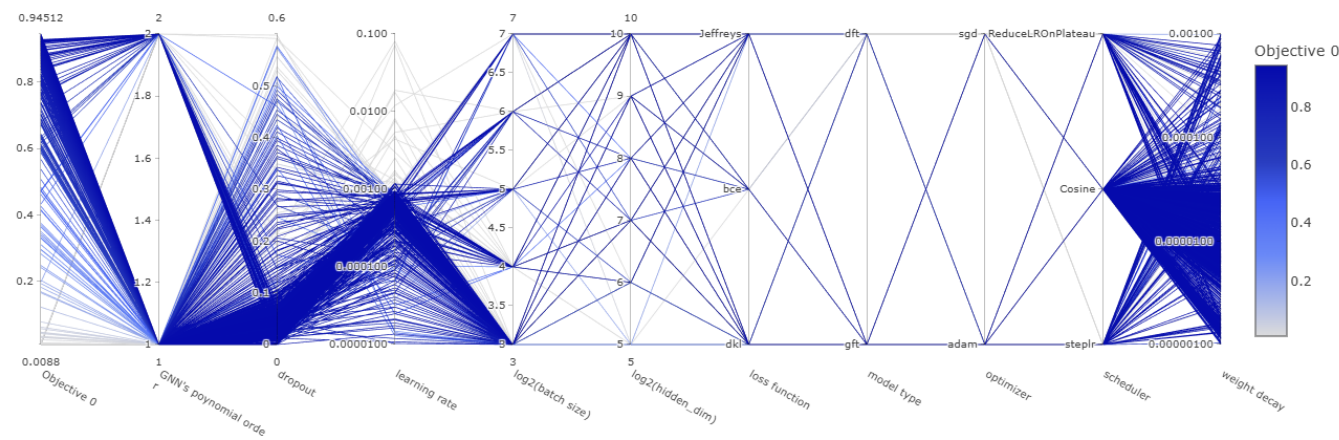


Preparing the Data for Learning

- **Split into 4 subsets:**
 - **70% Training**
 - **10% Validation** – hyperparameter tuning.
 - **10% Calibration** – conformal prediction.
 - **10% Test** – evaluation.
- **Consistency:** fixed random seed ensures splits are reproducible.

Training Process

- **Key hyperparameters tuned with Optuna (1000 trials):**
 - **The usual:** Dropout, hidden dimension, batch size, learning rate, weight decay
 - **Model type** (GFT vs. DFT)
 - **Loss function** (BCE, DKL, Jeffreys)
 - **Optimizer** (Adam, SGD)
 - **Polynomial order** (0, 1, 2)
 - **Scheduler** (StepLR, Cosine, ReduceLROnPlateau)



Results so Far

- Best results achieved for **GFT model** with **2 fully connected layers** and a **polynomial of order 1**.
- Managed to achieve **92.79%** accuracy for top 3.

Why so far?

Conformal Prediction

- **Goal:** Quantify prediction uncertainty in outage detection.
- **Key idea:** Instead of a single “best guess,” produce a **set of candidate outages**.
- **Guarantee:** With probability $\geq 1 - \alpha$ (e.g. 90%), the true outage(s) is in the set.



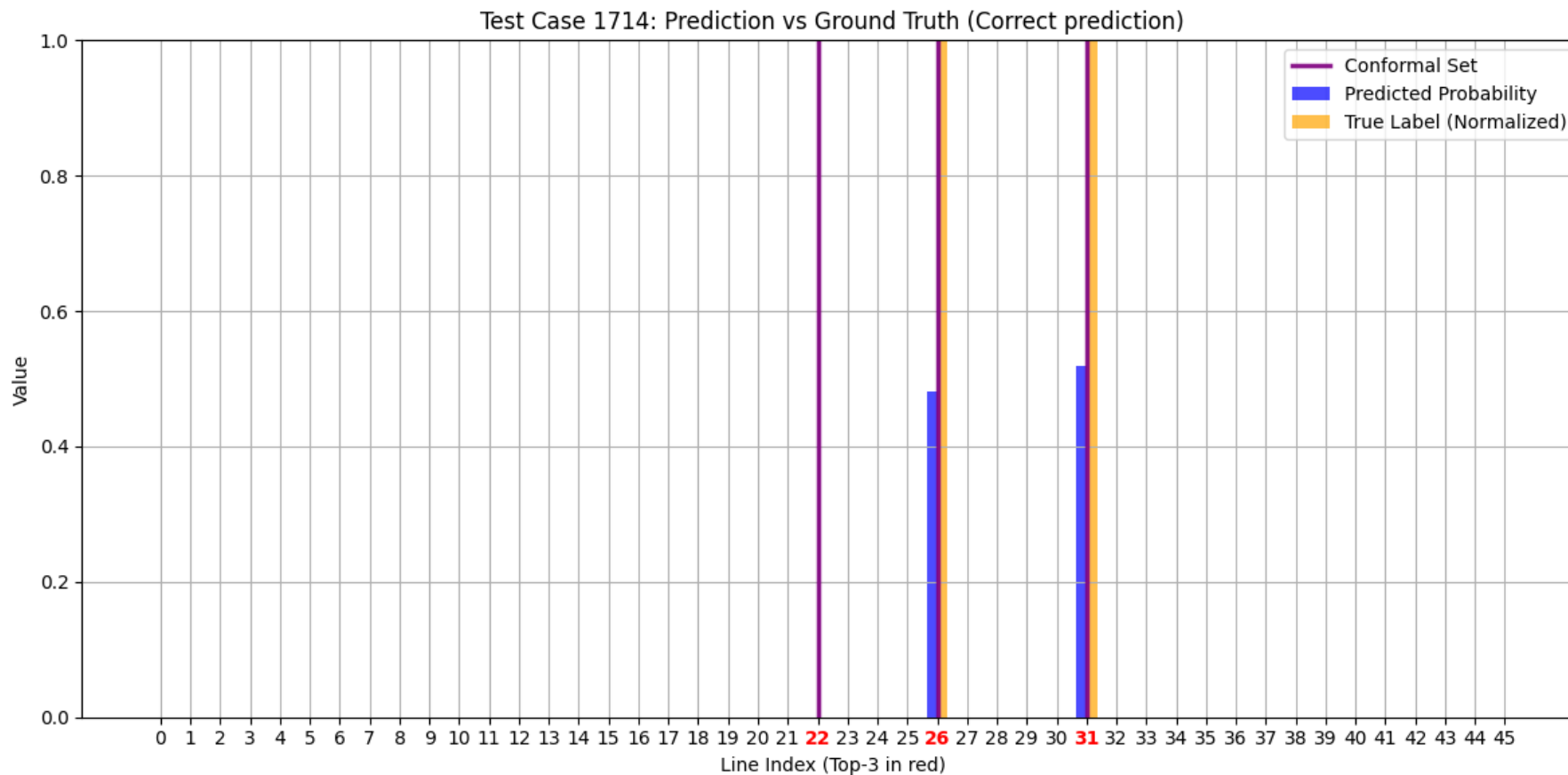
RAPS

- **RAPS** = Regularized Adaptive Prediction Sets.
- Developed by **Anastasios N. Angelopoulos** and **Stephen Bates**.
- Provides **valid coverage** while keeping the prediction sets small and practical.
- **In our case:** ensures the true outage is included in the set with high probability.

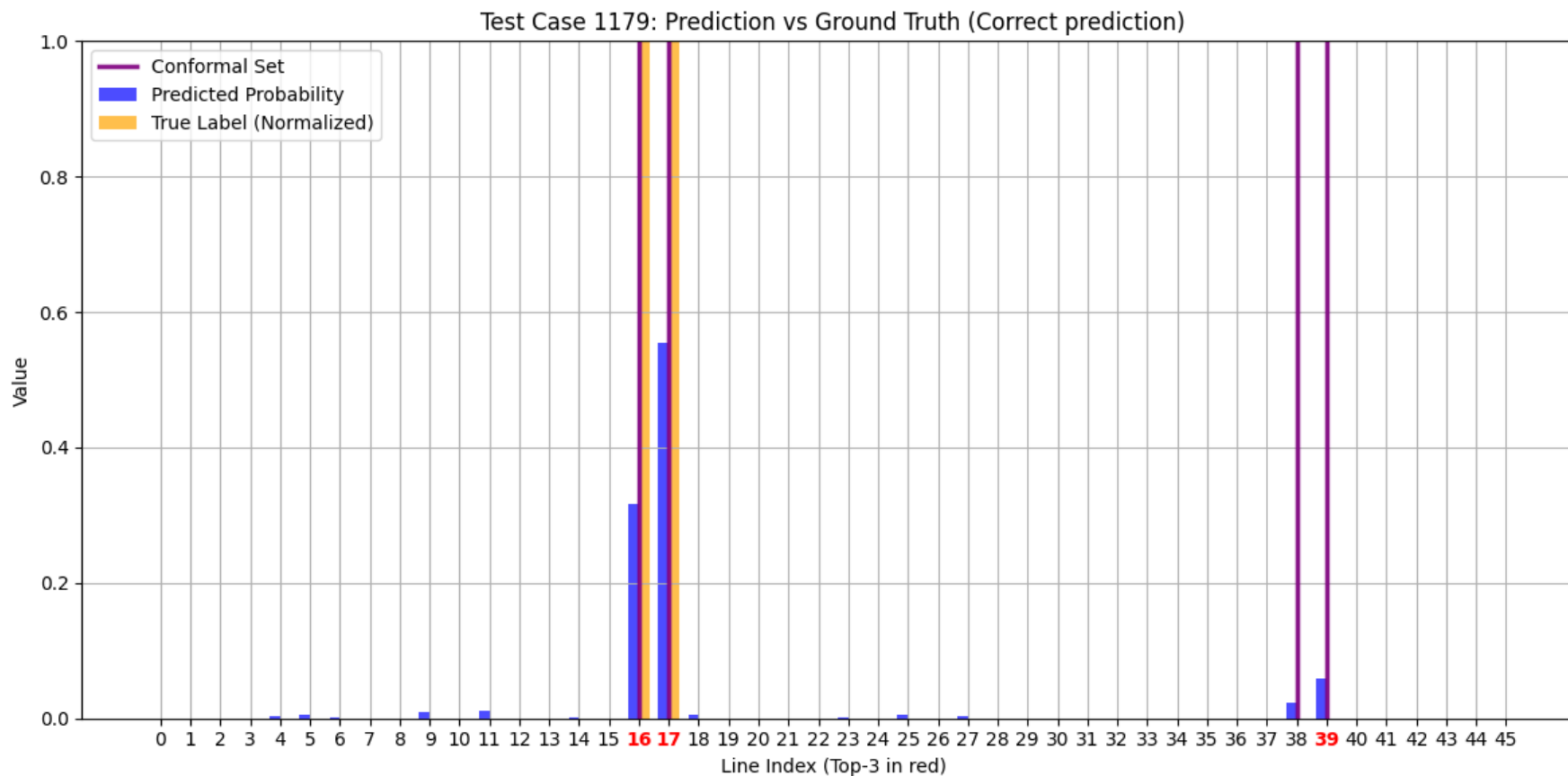
Results

- Using **RAPS** we managed to achieve **95.81%** accuracy with **average set size of 3.42**.
- more than 3% improvement for only 0.42 extra labels in the set on average!

Results



Results



Summary

- Leveraged the graph structure of the power grid to enable **fast and accurate outage detection**, even for multiple simultaneous outages.

How fast?

- GCN forward pass: $O(N^2)$ (~0.7 milliseconds for forward pass)
- Classical methods: $O(N^{3+k})$
- **Order on magnitude faster!**

Conclusions

- **Graph-based characteristics** of the power network enable effective detection of rare outage scenarios.
- Through CP, we achieve **statistical guarantees** on the black-box model's predictions, with **classical approaches still guiding the ultimate decision**.
- Complementary to Routtenberg et al. (Detection of False Data Injection Attacks in Smart Grids based on Graph Signal Processing), we **rely on PMU measurements** rather than outage data.

Future Work

- **Learnable adjacency weights** - letting the model tune edge strengths between buses
- **More reliable data** (or even real PMU data)
- **Random outage time data**
- **Test on larger/more complex grids**
- **Explore hybrid spatial + spectral GCN architectures**
- **Incorporate other measurements of the grid** (absolute voltage)

References

- Jia He, and Maggie Cheng. "Graph convolutional neural networks for power line outage identification."
- Anastasios N. Angelopoulos and Stephen Bates, "A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification."